DO NOW — RECALL

# Lesson 065 · Computational Methods: Recognising Solva

Name: _____ Class: _____ Date: _____

**1. What is a precondition?**

**2. What is abstraction?**

**3. How does a queue differ from a stack?**

**4. Trace binary search for 11 in the sorted, zero-indexed array [2,4,6,8,10,12,14]. Start low=0, high=6. Use mid=(low+high) DIV 2 and exclude the checked midpoint when updating bounds. State the checked indexes and the final result.**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

**1. Model answer**

Condition required for an algorithm to work correctly.

**2. Model answer**

Selecting relevant details and omitting irrelevant ones.

**3. Model answer**

FIFO versus LIFO.

**4. Model answer**

3, 5, 4; not found. Values are 8, 12, 10. The final bounds are low=5, high=4, so the search stops.

**By the end, you can…**

• A solvable computational problem needs an adequately precise input representation, a finite method and a verifiable output or objective.

• Problem recognition identifies the class of task—classification, searching, route finding, scheduling or optimisation—before selecting a method.

• Define measurable success criteria and constraints; vague goals such as 'best route' require clarification of 'best' (distance, cost or time).

**Vocabulary — know these terms**

problem recognition — Identify the underlying task type.

constraint — Requirement a solution must obey.

objective function — Quantity to minimise or maximise.

computable specification — Precise task that a program can implement/check.

NEW LEARNING

# Read, annotate and explain

### Identify

A solvable computational problem needs an adequately precise input representation, a finite method and a verifiable output or objective.

### Model

Problem recognition identifies the class of task—classification, searching, route finding, scheduling or optimisation—before selecting a method.

### Evaluate

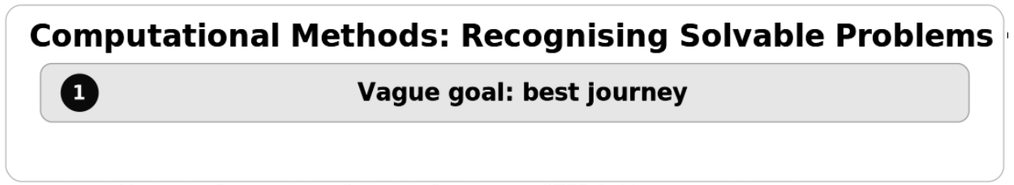
Define measurable success criteria and constraints; vague goals such as 'best route' require clarification of 'best' (distance, cost or time).

**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**New Learning**




Computational Methods: Recognising Solvable Problems · STEP 1: Vague goal: best journey

**New Learning**




Computational Methods: Recognising Solvable Problems · STEP 2: Vague goal: best journey; Define start and goal

**New Learning**




Computational Methods: Recognising Solvable Problems · STEP 3: Vague goal: best journey; Define start and goal; Choose edge metric: cost



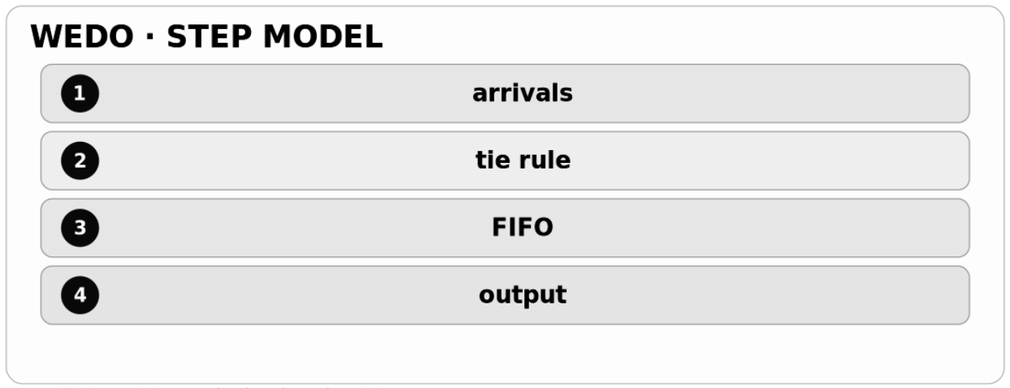


# We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Turn 'pick the fairest lunch queue order' into a computable FIFO rule.**

An ordering rule makes fairness measurable and repeatable within the chosen model.




WEDO · STEP MODEL: arrivals; tie rule; FIFO; output

WE DO — CHECK AND IMPROVE

# Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

### Guided 1 — CHECK

Explain the decision
An ordering rule makes fairness measurable and repeatable within the chosen model.

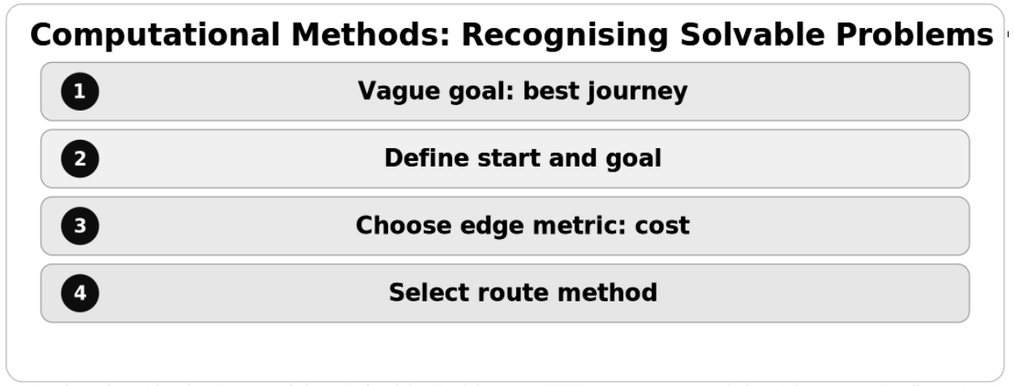




# New Learning · original lesson diagrams

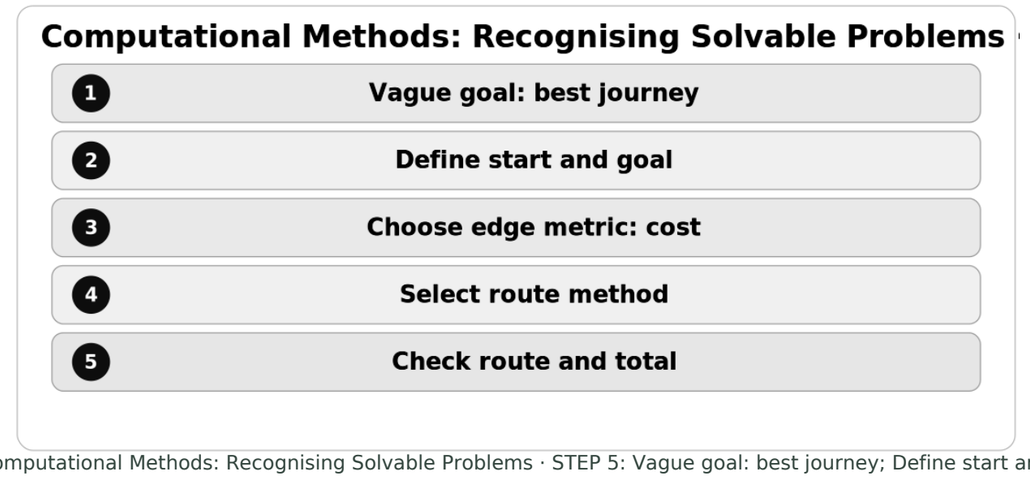
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**New Learning**




Computational Methods: Recognising Solvable Problems · STEP 4: Vague goal: best journey; Define start and goal; Choose edge metric: cost; Select route method

**New Learning**




Computational Methods: Recognising Solvable Problems · STEP 5: Vague goal: best journey; Define start and goal; Choose edge metric: cost; Select route method; Check route and total

I DO — WORKED EXAMPLE

# Follow the worked method

**Turn 'choose the best bus route' into a precise computational problem.**

**Explain the decision**

Cost is defined and connections must be valid; output stops and total fare.





EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking.

**Every vague human objective has exactly one obvious numerical optimisation target.**

**Circle: Confident / Getting there / Need help**

Explain what you can now do independently.

**GREEN PEN — Correct and explain**

Choose one error. Write the correct method and explain precisely why your first response was wrong.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

**Question 3**

Evaluate whether a computer can choose a 'good teacher' simply from exam results, explaining models, constraints and limits. [6]





# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Turn 'choose the best bus route' into a precise computational problem.**

Cost is defined and connections must be valid; output stops and total fare.




IDO · STEP MODEL: start; dest; fares; minimum cost

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

### Guided 1 — TRY

Turn 'pick the fairest lunch queue order' into a computable FIFO rule.
Source code / input:
Input ordered arrival timestamps and ticket IDs; output same order, oldest first; handle ties by sequence number.





YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

### Question 1

Define computational problem input, process, output for cheapest route on a network. [6]

### Question 2

A pupil asks to 'sort books by best'. Give two precise orderings, explain why different and select one for finding a title. [6]